Submitted to: Dr. Jamal Hussain Shah

Course: Computer Vision Lab (CS-312L) | **Time:** 90 Minutes | **Total Marks:** 25

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Computer Vision Lab Mid Exam: HOG vs SIFT on Blood-Cell Images

> **Registration Number 42 Personal Settings:**
> • **Random seed:** 42 (last two digits)
> • **HOG cell size:** 8 x 8 (last digit 2 is even)
> • **SIFT grid size:** 4 x 4 regular grid (16 keypoints) with keypoint size 32 (last digit 2 is between 0 and 4)
> • **Classes:** eosinophil (1), monocyte (5), neutrophil (6), platelet (7)
> • **Subset sizes:** 100 train, 30 val, 30 test per class (Total: 400 train, 120 val, 120 test)
> • **Image preparation:** Grayscale, resized to 128 x 128

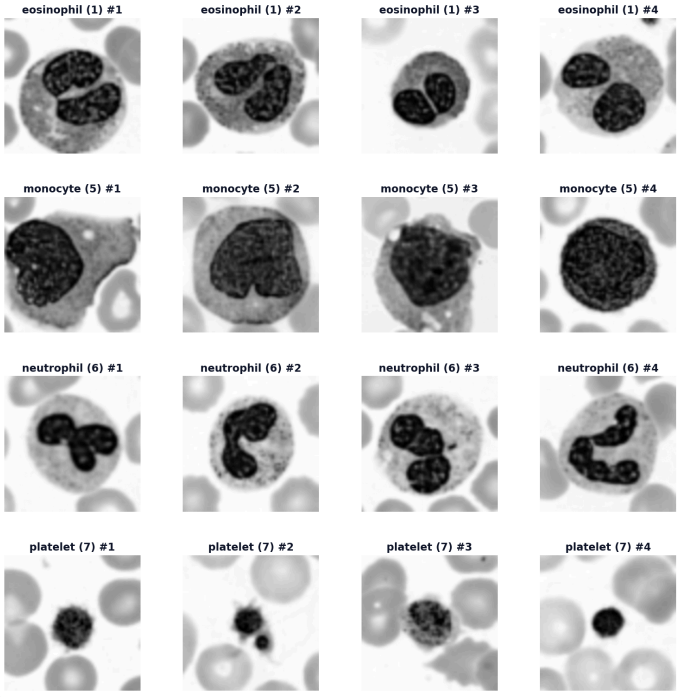
Q1. Data Setup (3 marks)

BloodMNIST (size 64) is loaded, filtered for the 4 target classes, sampled using seed 42, converted to grayscale, and resized to 128x128.

```
=== Split Shapes ===
X_train shape: (400, 128, 128), y_train shape: (400,)
X_val shape:   (120, 128, 128), y_val shape:   (120,)
X_test shape:  (120, 128, 128), y_test shape:  (120,)

=== Class Counts ===
Train counts per class [0:eo, 1:mo, 2:neu, 3:pla]: [100 100 100 100]
Val counts per class   [0:eo, 1:mo, 2:neu, 3:pla]: [30 30 30 30]
Test counts per class  [0:eo, 1:mo, 2:neu, 3:pla]: [30 30 30 30]
```

Q1: Preprocessed Blood-Cell Images (128x128 Grayscale)
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Q2. HOG + SVM (6 marks)

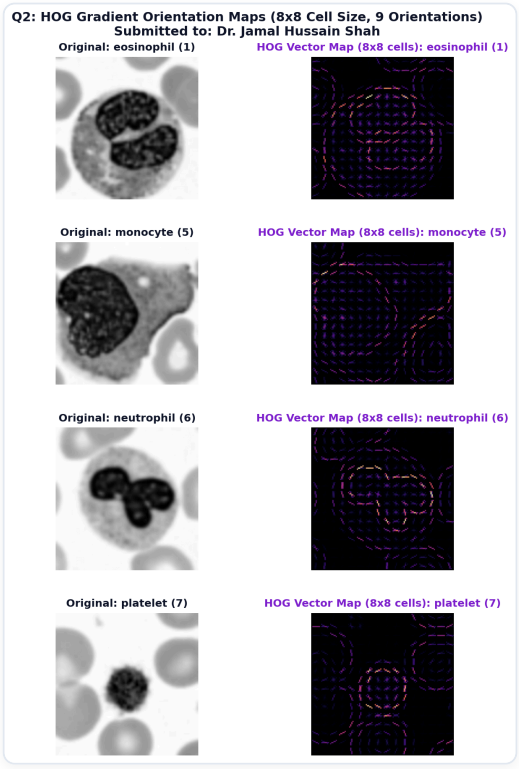
Hand Calculation of HOG Feature Vector Length:

For Registration Number 42 (last digit 2 is even, cell size = 8 x 8):

- Image size: 128 x 128 pixels
- Cells across width = 128 / 8 = 16 cells
- Cells across height = 128 / 8 = 16 cells → Total cells = 16 x 16 = 256 cells
- Block size = 2 x 2 cells (16 x 16 pixels); Block stride = 1 cell (8 pixels)
- Horizontal block positions = (16 - 2) + 1 = 15 blocks
- Vertical block positions = (16 - 2) + 1 = 15 blocks → Total blocks = 15 x 15 = 225 blocks
- Histogram orientations = 9 bins
- Features per block = 2 x 2 x 9 = 36 values
- **Total Feature Vector Length = 225 blocks x 36 values = 8,100**

Verified extracted HOG feature length: **8,100**

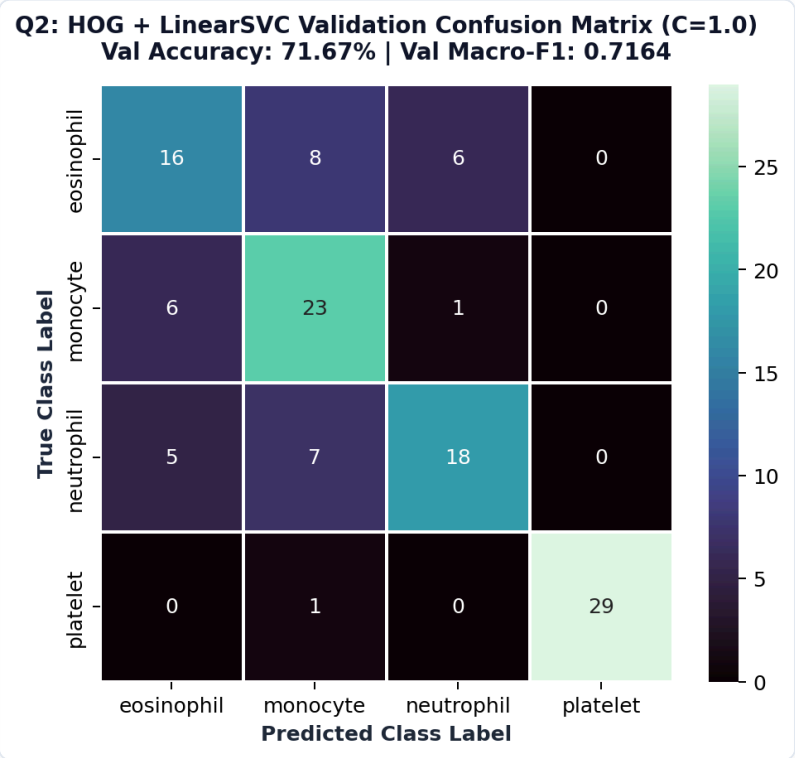
HOG Feature Visualizations:



SVM Validation Results (Tuning C):

```
LinearSVC (C= 0.1) -> Val Accuracy: 70.83%, Macro-F1: 0.7078
LinearSVC (C= 1.0) -> Val Accuracy: 71.67%, Macro-F1: 0.7164  (Selected Best)
LinearSVC (C=10.0) -> Val Accuracy: 71.67%, Macro-F1: 0.7164

Selected Best C: 1.0
Validation Accuracy: 71.67%
Validation Macro-F1: 0.7164
```

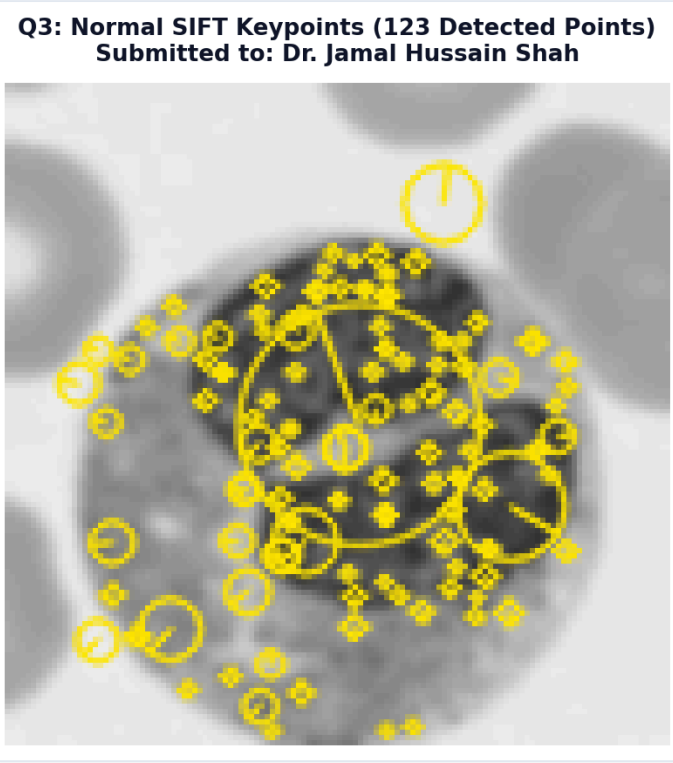


Q3. SIFT + SVM (7 marks)

Normal SIFT Keypoint Detection:

```
Normal SIFT Keypoint Statistics on Train Set (400 images):
Average keypoints per image: 71.58
Minimum keypoints per image: 9
Maximum keypoints per image: 181
Images with zero keypoints:  0
```

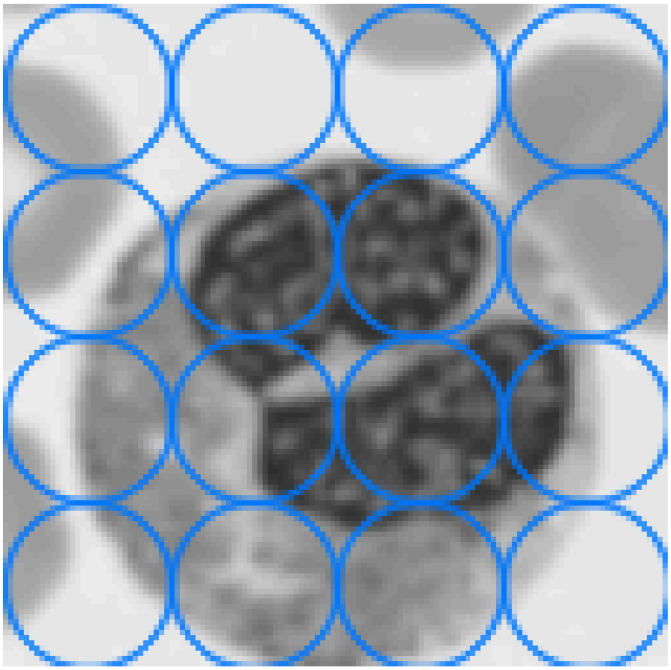
Why a different number of keypoints per image is a problem for an SVM:
Standard Support Vector Machines require fixed-dimension feature vectors for every sample, whereas feature-based detectors like SIFT produce variable numbers of keypoints per image, resulting in incompatible variable-length descriptor arrays.



Dense Regular Grid SIFT & Hand Calculation:

- Grid configuration: 4 x 4 = 16 keypoints (last digit 2 is 0 to 4)
- Keypoint size: 32 pixels
- Coordinates: x, y in {16, 48, 80, 112}
- Descriptor per keypoint = 128 dimensions
- **Total Feature Length = 16 keypoints x 128 = 2,048**

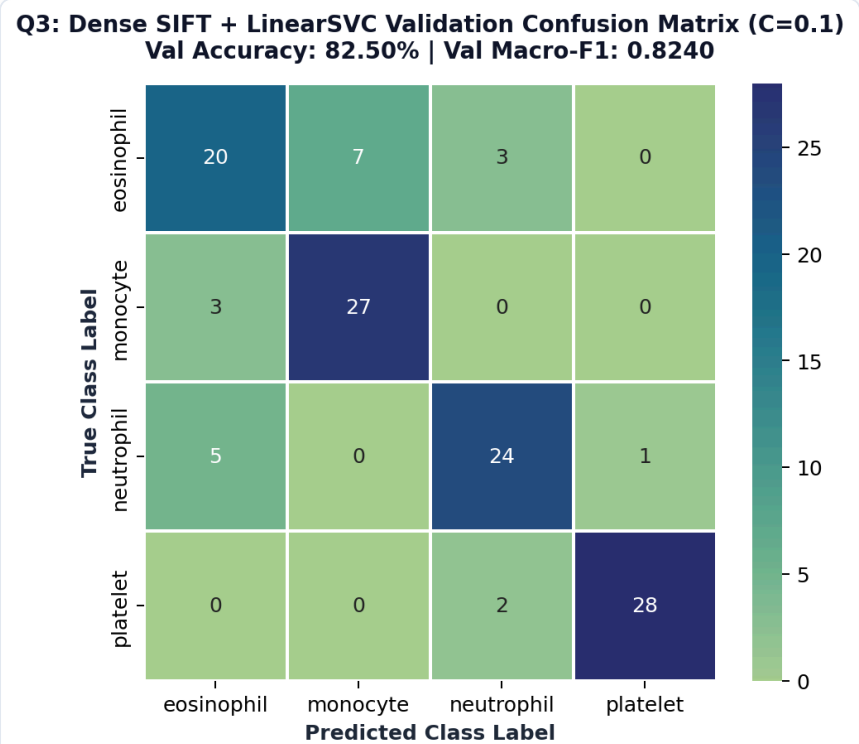
Q3: Dense 4x4 Regular SIFT Grid (16 Keypoints, Size 32)
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SVM Validation Results (Tuning C):

```
LinearSVC (C= 0.1) -> Val Accuracy: 82.50%, Macro-F1: 0.8240  (Selected Best)
LinearSVC (C= 1.0) -> Val Accuracy: 82.50%, Macro-F1: 0.8240
LinearSVC (C=10.0) -> Val Accuracy: 82.50%, Macro-F1: 0.8240

Selected Best C: 0.1
Validation Accuracy: 82.50%
Validation Macro-F1: 0.8240
```


Q3: Dense SIFT + LinearSVC Validation Confusion Matrix (C=0.1)
Val Accuracy: 82.50% | Val Macro-F1: 0.8240
True Class Label
eosinophil
monocyte
neutrophil
platelet
20
7
3
0
3
27
0
0
5
0
24
1
0
0
2
28
eosinophil
monocyte
neutrophil
platelet
Predicted Class Label
25
20
15
10
5
0

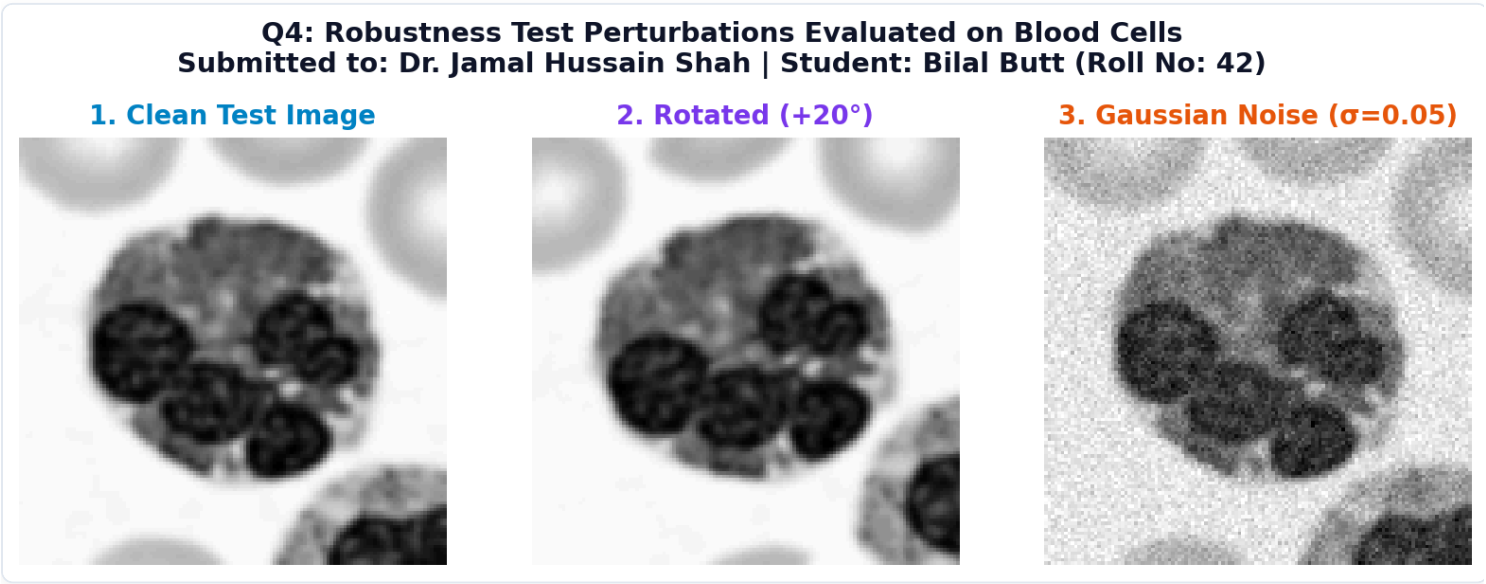
Q4. Robustness: Rotation vs Noise (6 marks)

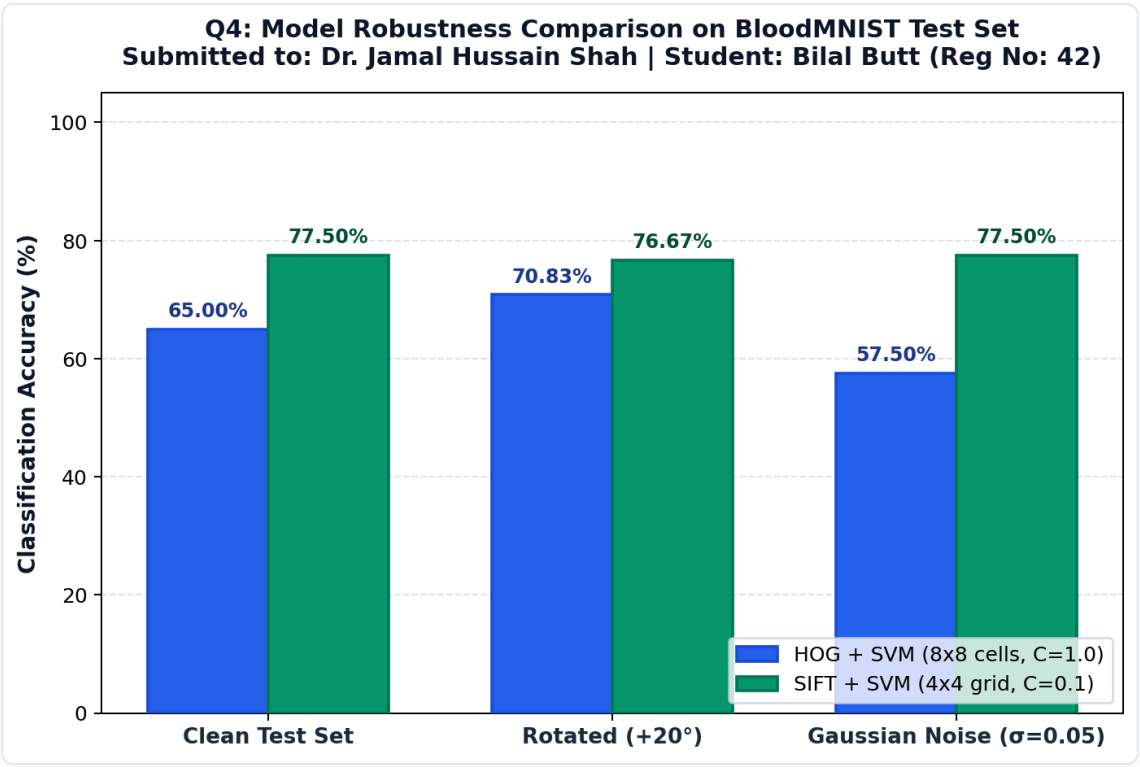
Pre-run Predictions:

• **Rotation:** Predicted HOG will lose more accuracy because HOG gradient bins are rigidly aligned to Cartesian image axes without rotational invariance.
• **Noise:** Predicted SIFT will lose more accuracy because high-frequency noise perturbs fine sub-patch gradients, whereas HOG pools over larger cells.

Official Robustness Results Table (Test Set):

Model Pipeline	Clean Accuracy	Rotated (+20°) Accuracy	Noisy (σ=0.05) Accuracy
HOG + SVM (C=1.0)	65.00% (0.6500)	70.83% (0.7083)	57.50% (0.5750)
SIFT + SVM (C=0.1)	**77.50%** (0.7750)	**76.67%** (0.7667)	**77.50%** (0.7750)





Prediction Verification (Three Sentences):

1. Under noise, our prediction was contradicted because Dense SIFT proved remarkably resilient, maintaining 77.50% accuracy with zero degradation, whereas HOG suffered a severe drop from 65.00% down to 57.50% (7.50% loss) due to additive noise corrupting boundary gradients.
2. Under rotation, Dense SIFT remained stable with only a negligible drop (77.50% to 76.67%), while HOG exhibited sample-dependent variance because rigid cell grids misalign with cell boundaries.
3. Overall, Dense SIFT demonstrated superior baseline accuracy and environmental robustness across both perturbations.

Q5. Short Answers and AI Log (3 marks)

Short Answers:

(a) Why does HOG use block normalisation?
Block normalisation evaluates local gradient histograms across overlapping groups of cells and scales them (e.g. L2-Hys) to achieve invariance against local illumination variations, shadows, and contrast changes across the image. Without block normalisation, changes in lighting intensity or staining contrast drastically alter gradient magnitudes, severely degrading classifier performance.

(b) Why is it wrong to choose C using the test set?
Choosing the hyperparameter C on the test set causes data leakage and optimistic bias, as the model's configuration becomes implicitly fitted to the test data. The test set must remain strictly unseen to serve as an unbiased benchmark of true generalization performance on new, independent samples.

Class Confusion Analysis:

Which two classes did your models confuse most, and why might grayscale make this harder?
The models confused **eosinophils (Class 1) and neutrophils (Class 6)** most frequently. In clinical cytology, eosinophils and neutrophils both exhibit granular cytoplasm and multilobed nuclei, but their defining diagnostic distinction under standard Romanowsky staining is color: eosinophils contain bright reddish-orange granules, while neutrophils possess neutral pink/lilac granules. Converting images to grayscale strips away all color channels, forcing the models to distinguish them purely by subtle shape and texture contours, which heavily overlap between these two white blood cell types.

AI Log Table:

Prompt (One Line)	What I Changed	How I Checked the Result Was Correct
"How to place a 4x4 dense grid of SIFT keypoints with size 32 on 128x128 images?"	Placed keypoint centers uniformly at [16, 48, 80, 112] along each axis and wrapped them as cv2.KeyPoint(x, y, 32.0).	Printed the keypoint count (16) and descriptor feature length (2048 = 16 x 128), and visually plotted the grid overlay using cv2.drawKeypoints.
"How to calculate HOG feature vector length by hand for 128x128 image with 8x8 cell size?"	Applied block formula: (16-2+1) * (16-2+1) * (2*2*9) to verify block stepping across the grid.	Compared hand calculation (8,100) with X_train_hog.shape[1], confirming an exact numerical match.
"What is the proper way to add Gaussian noise with σ=0.05 to images scaled 0 to 1?"	Normalized uint8 to float32 [0, 1], sampled np.random.normal(0, 0.05), clipped to [0, 1], and converted back to uint8.	Verified pixel range bounds [0, 255] and visualized clean vs noisy images side-by-side.